

NCMAS 2020

and how to access other HPC, Cloud

Presented by

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Why are we here



National Computational Merit Allocation Scheme (NCMAS)

“... provides *access*, based on *research and computational merit*, for researchers at Australian universities and publicly funded research agencies, *to major national HPC facilities*.”

Supported by Commonwealth Government NCRIS and Super Science programs.

NCMAS 2020 HPC Facilities



115+ MSUs

New HPC!



100 MSUs



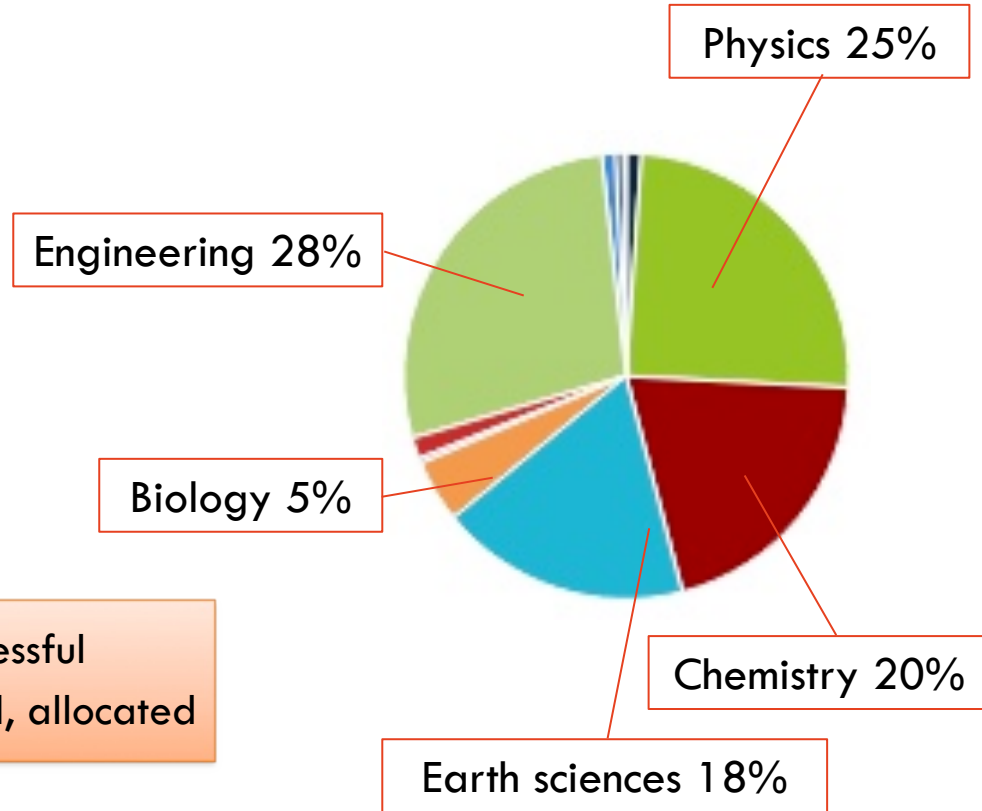
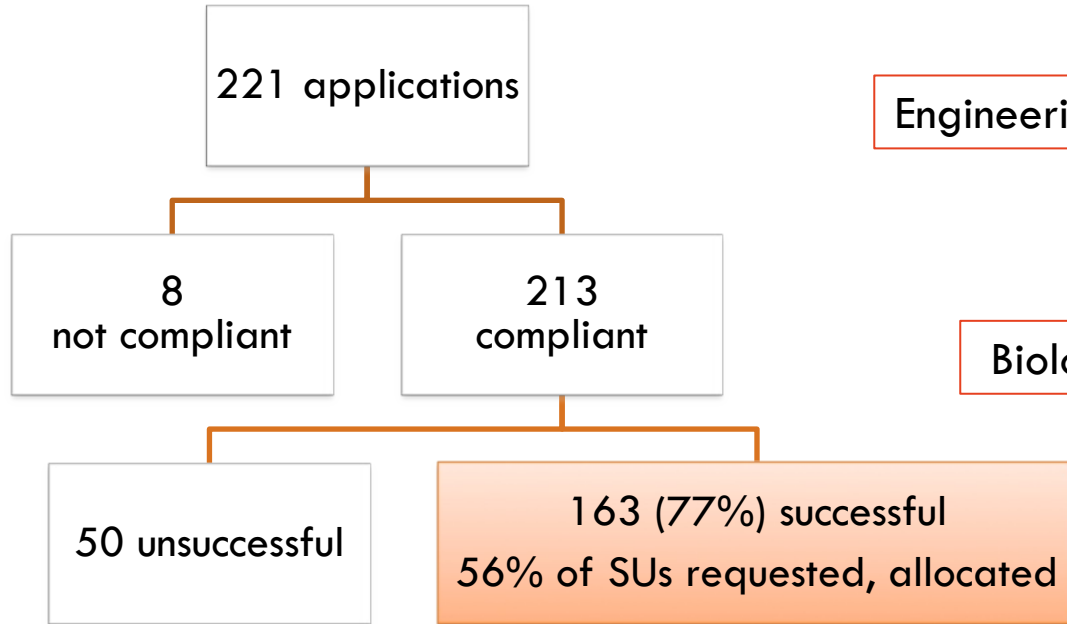
2.5 MSUs



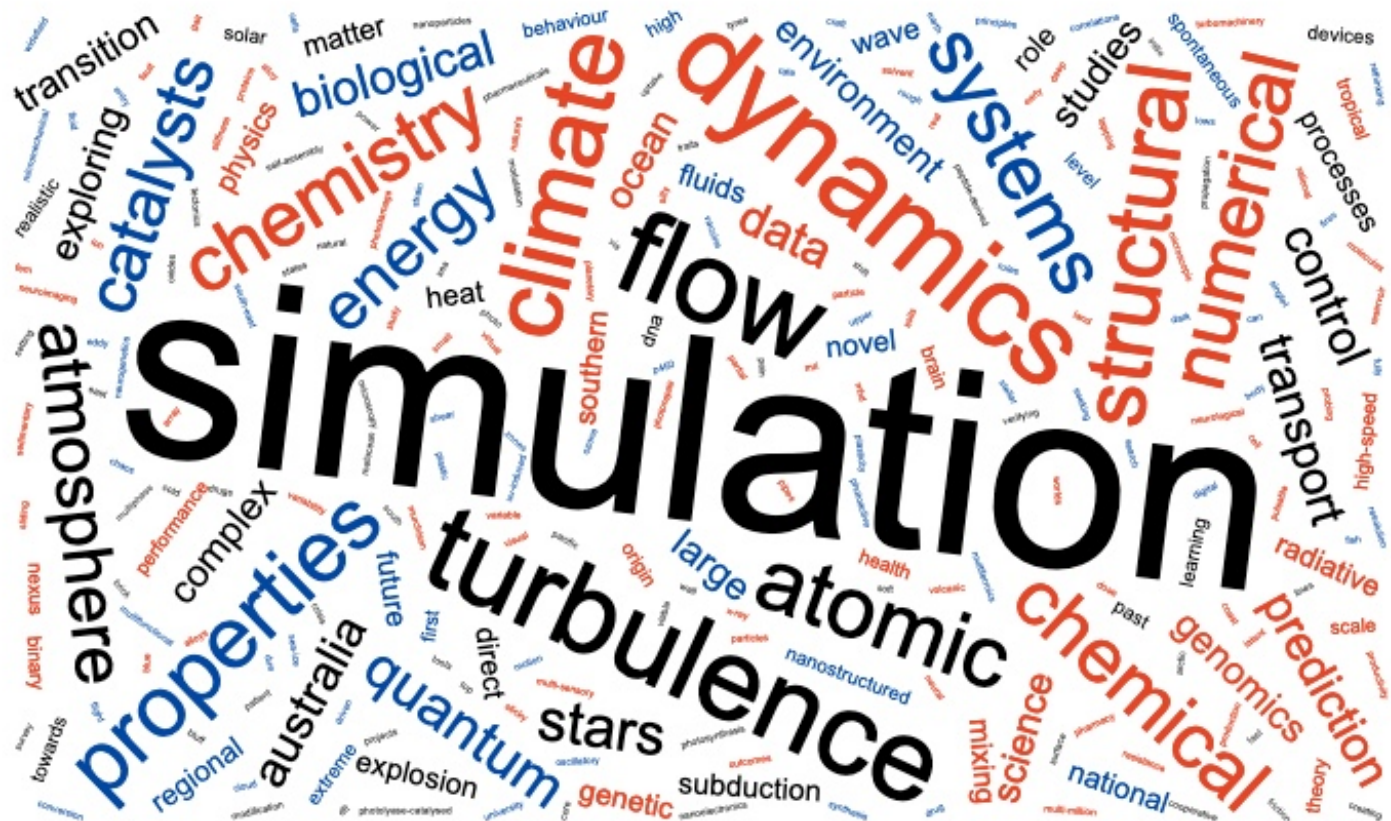
THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA **FlashLite**

1.93 MSUs

2019 Success Rate



Successful projects



Key Dates

Now

- Run pilot studies on *Startup/Director* shares – SIH can help!

2 Aug 2019

- Applications open

20 Sep

- Applications close

23 Sep – 4 Oct

- Admin and Tech assessment

8 Oct – 22 Nov

- Merit assessment

25-26 Nov

- Allocation committee meeting

4 Dec

- Release of outcomes

Eligibility

Lead CI

- Based in Australia
- Hold substantive position at an Australian university, research institute, or Government science agency (0.2 FTE or greater)
- Has a track record of research funding/grants
- Can apply for *one* NCMAS application only

ECRs

- Has *independent funding support* (e.g. DECRA, USyd Fellowship)
- PhD conferred within last 5 years

PhD students not eligible – Ask your supervisor to apply!

NCMAS Categories

- Open/Unrestricted
 - Requests for > 1 MSU assessed most rigorously by entire committee
- Early Career Researcher
 - 250kSU at one facility
- Special Consideration (e.g. career interruption)

Writing a competitive application

1. Project significance

- Scientific merit, not operational workload
- Grant support for project e.g. ARC, NHMRC

2. Effective use of HPC *at scale*

Demonstrate use of:

- Parallel jobs, efficient multi-CPU jobs
- Data interconnects
- High-performance storage

or:

- Embarrassingly parallel workflows *at scale*
- Dependency in large-scale or national research data collections

Writing a competitive application cont.

3. Past HPC experience

- Cite publications that acknowledge HPC
- Describe HPC facilities used – including the facility you are applying for
- Experience with other allocation schemes e.g. Startup/Partner/Director schemes

4. Accurate estimation of resources needed

- Comprehensive justification for resources requested
- Evidence of *your* data and workflows on the facility you are applying for

Here is a real-world example using NWChem

Cores	Nodes	Walltime (hours)	Cost (Node hours)
128	8	11.2	90
256	16	4.4	70
512	32	2.1	67
1024	64	1.0	64
2048	128	0.91	116
4096	256	0.75	192

Ideal – efficient core hours and walltime

How to apply

1. New users and projects:

- Register for a new NCI user account
- Submit a project proposal (selecting the NCMAS resource scheme) at <https://my.nci.org.au>

2. Existing projects:

- Update project details, your profile, and project personnel in the NCI registration system <https://my.nci.org.au>

3. All: Complete the full NCMAS application for the project at <https://ncmas.nci.org.au>

Other HPC merit schemes

- Pawsey Energy and Resources Scheme
 - Opens 2nd August
 - Min 250 kSUs per project
- Intersect Compute Merit Allocation Scheme (ICMAS)
 - Opens Q4 2019
- Intersect pilot projects (<10kSU)
 - All year

Where eligible apply for all NCMAS, ICMAS, and Pawsey ERS

Other HPC computing resources



Artemis HPC

- 7636 CPUs, 108 NVIDIA V100 GPUs, 56 Gbps Infiniband networking, Lustre filesystem
- Access: [dashR](#)

Other Virtual Machines



Argus Virtual Research Desktops (VRDs)

- Interactive visualization and graphical processing
- Up to 16 vCPUs, 8vGPUs, 256GB RAM
- Access: sih.info@sydney.edu.au



ICT VMs

- Up to 2 CPUs, 2GB RAM, 50GB disk
- Hosting websites, surveys
- Access: <https://personalvm.sydney.edu.au>



90 Day VM

Personal Temporary VM

Request

Other Virtual Machines



OwnTime VMs

- Up to 64 CPUs, 1TB RAM
- Access: [Intersect helpdesk](#) (and approved by USyd)



Pawsey Nimbus Cloud

- Up to 64 CPUs, GPUs, 384GB RAM
- Access: <https://apply.pawsey.org.au>



NCI Tenjin Cloud

- Up to 16 CPUs, 244GB RAM, 640GB disk
- Access: enquiries@nci.org.au

Other Commercial Cloud

Access:

- Free Tier
- \$\$
- Research credits:



AI for Earth, Good, Accessibility, Humanitarian Action

Other projects: Auda Eltahla a-aelta@microsoft.com



Cloud credits for Research



GCP Research Credits

More information



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